TEST CODE **01207020**

FORM TP 2007002

JANUARY 2007

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

$1\frac{1}{2}$ hours

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **There are FIVE questions in this booklet. Answer ALL questions.**
2. **You MUST use this answer booklet when responding to the questions. For EACH question, write your answer in the space provided and return the answer booklet at the end of the examination.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Copyright © 2006 Caribbean Examinations Council®.
All rights reserved.

Answer ALL questions.

DO NOT spend more than 30 minutes on Question 1.

1. The table below shows the changes in height of a corn plant over a 36-day period.

TABLE 1: CHANGES IN HEIGHT OF A PLANT OVER A 36-DAY PERIOD

Time/days	0	6	12	18	24	30	36
Height/cm	0	10	90	215	245	250	250

(a) (i) On the grid provided in Figure 1, draw a graph of the results shown in Table 1.

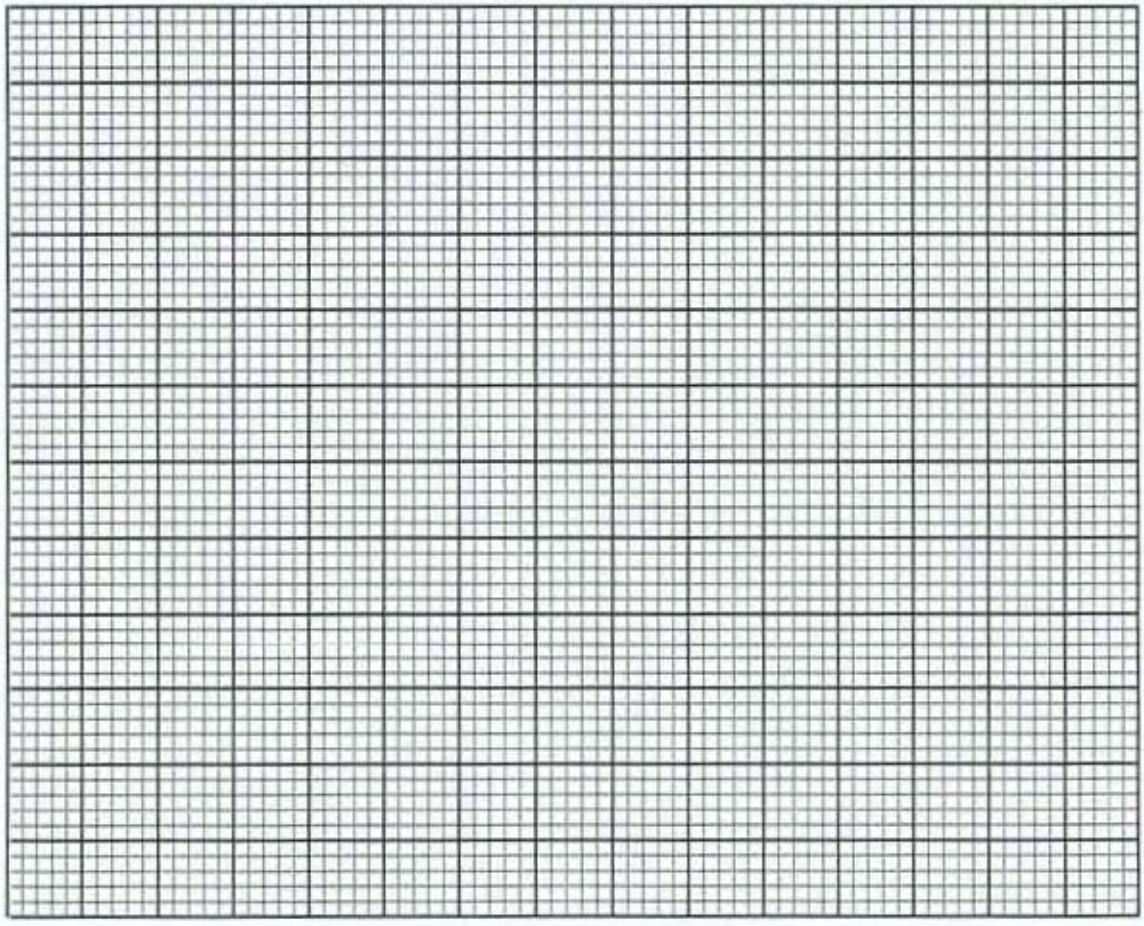

Figure 1. Grid for graph of results shown in Table 1

(7 marks)

(ii) Using information from the graph you constructed on page 2, estimate the height of the corn plant on Days 5 and 13.

Day 5: _____

Day 13: _____

(2 marks)

(b) Describe the pattern of growth shown by the plant over the 36-day period.

_____

_____

(3 marks)

(c) (i) What is the likely source of food for the corn plant during the first 6 days?

_____

(1 mark)

(ii) Give TWO uses of the food obtained by the corn plant during the first 6 days.

_____

_____

(2 marks)

(d) Another method of determining growth of the plant is using the dry mass (weight).

(i) Describe the method for obtaining dry mass, including the materials and apparatus necessary.

Materials and apparatus:

_____

_____

(2 marks)

Method:

_____

_____

_____

_____

(2 marks)

(ii) State ONE step to be taken to ensure the accuracy of the results when using this method.

_____

_____

(1 mark)

(iii) Give ONE disadvantage of using dry mass to determine growth.

_____

_____

(1 mark)

(e) (i) Suggest TWO ways in which the growth of the corn plant would be different over the 36 days, if the soil in which it grew were deficient in nitrogen and magnesium.

_____

_____

_____

(2 marks)

(ii) Explain how a lack of EACH of the following elements may result in the differences suggested in (e)(i) above.

Nitrogen: _____

_____

Magnesium: _____

_____

(2 marks)

(f) (i) The corn seed germinated after a heavy shower of rain, although it had been in the soil for a long period of time. What changes occurred in the seed for germination to take place?

_____

_____

_____

(3 marks)

(ii) Apart from water, state TWO conditions necessary for germination.

_____

_____

(2 marks)

Total 30 marks

2. Figure 2 shows the incidence of osteoporosis, a bone disease in humans.

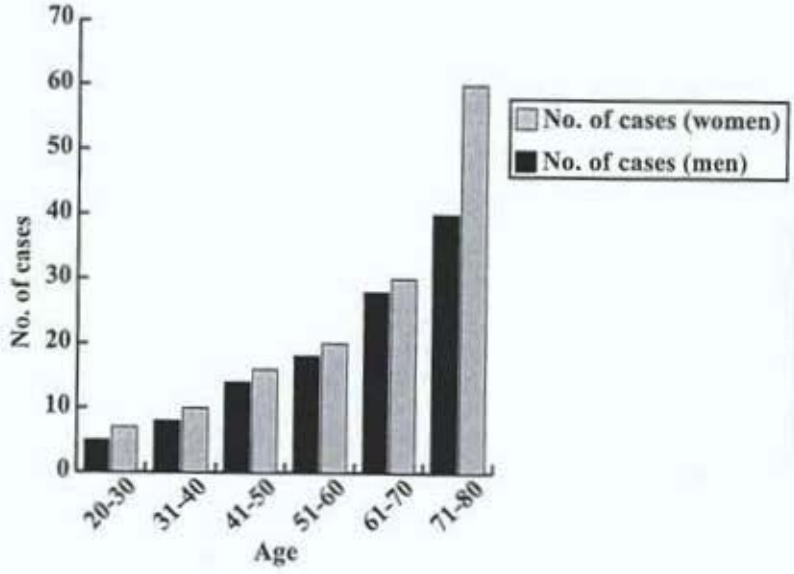

Figure 2. Incidence of osteoporosis in males and females.

(a) Using the information in Figure 2, draw TWO conclusions about the incidence of the disease.

_____

_____

(2 marks)

GO ON TO THE NEXT PAGE

(b) Osteoporosis refers to a decrease in bone density, the bones become brittle, weakened and easily fractured.

(i) Give TWO reasons that may account for the development of osteoporosis with age.

_____

_____

(2 marks)

(ii) It has been suggested that osteoporosis is associated with oestrogen levels in women.

State TWO functions of oestrogen.

_____

_____

(2 marks)

(c) (i) If children were to be affected by osteoporosis, suggest TWO likely effects of the disease on their bodies.

_____

_____

(2 marks)

(ii) Suggest a mineral supplement that is likely to be recommended for people suffering from osteoporosis.

_____

(1 mark)

(d) (i) Healthy bone is living tissue. Give TWO advantages of bone being a living tissue.

_____

_____

(2 marks)

(ii) Apart from support, identify TWO functions of bone.

_____

_____

(2 marks)

Total 13 marks

3. Figure 3 shows the menstrual cycle in a human female.

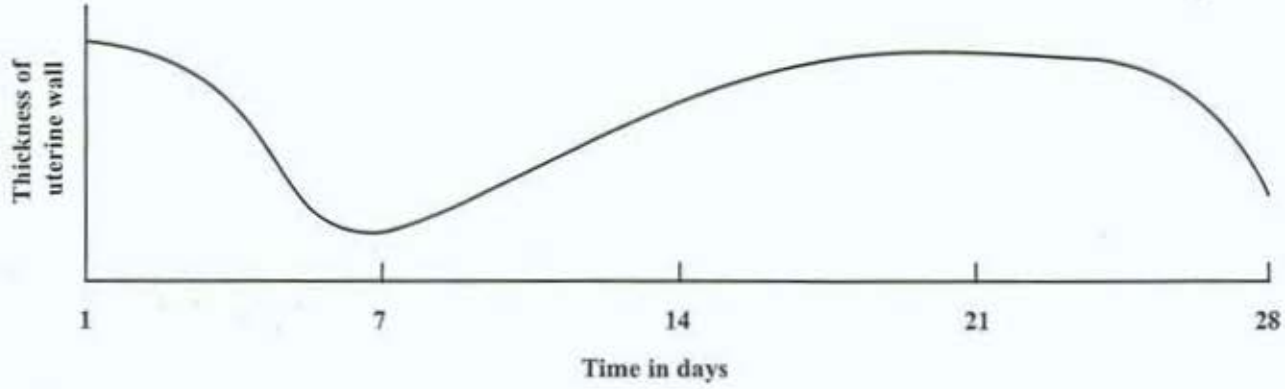

Figure 3. Menstrual cycle of the human female

(a) (i) Using the numbers I - IV, label Figure 3 to indicate where in the menstrual cycle EACH of the following events occur:

a) Uterus wall builds up as oestrogen level rises (I)

b) Many blood vessels develop (II)

c) Shedding of the uterine wall (III)

d) Uterine wall prepared for implantation (IV) **(4 marks)**

(ii) Identify TWO hormones involved in the menstrual cycle and give the function of any ONE.

_____

_____

_____

(3 marks)

(b) State THREE functions of the placenta during development of the foetus.

_____

_____

_____

(3 marks)

(c) (i) State THREE birth control methods practised by human beings.

_____

_____

_____

(3 marks)

(ii) Suggest why birth control seems more important in humans than in other populations of mammals.

_____

_____

_____

(2 marks)

Total 15 marks

4. Figure 4 below represents one cell from a plant and one from an animal.

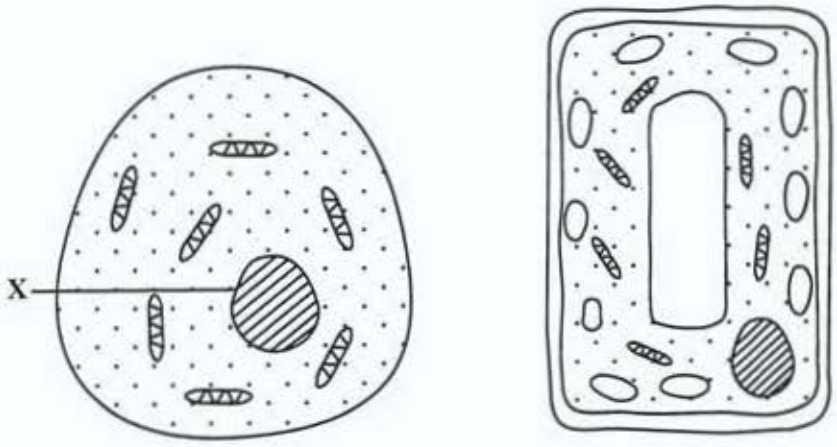

Figure 4. Plant and animal cells as seen under a microscope

(a) (i) Using only the letters P, Q and R label the structures on both cells that perform EACH of the following:

a) Produce ATP (P)
b) Synthesise protein (Q)
c) Allow substances to enter and leave the cell (R) **(3 marks)**

(ii) Using the letters S and T, label the structure on Cell 2 ONLY that performs EACH of the following functions:

a) Converts light into chemical energy (S)
b) Allows a wide range of substances to pass through (T) **(2 marks)**

(b) (i) Name structure X shown in Cell 1.

_____

(1 mark)

(ii) Structure X is absent from mature red blood cells and sieve tubes.
Suggest one process which these cells will not be able to perform. Give a reason for your answer.

Process:

_____

Reason:

_____

(2 marks)

(c) (i) What is likely to occur in Cells 1 and 2 of Figure 4 if they were placed in distilled water for over half-an-hour?

Cell 1:

_____

_____

Cell 2:

_____

_____

(4 marks)

(ii) Explain why the changes in Cell 1 might be different from the changes in Cell 2 after half-an-hour in distilled water.

_____

_____

(2 marks)

(d) Although blood cells are in a watery medium (plasma) they do not absorb excessive amounts of water. Explain why this is so.

_____

_____

(2 marks)

(e) Persons living at very high altitudes where the air is rare (low oxygen concentration) have a larger number of red blood cells per unit volume of blood than those living at sea level.

Suggest why a person living at high altitudes requires a relatively large number of red blood cells.

_____

_____

(2 marks)

Total 18 marks

5. Figure 5 is a labelled diagram of part of the breathing apparatus of fish.

Figure 5. Fish gill

(a) (i) Water passes over the gill rakers before passing over the filaments. Suggest the function of the gill rakers.

_____

(1 mark)

(ii) Identify ONE feature observed in Figure 5 that allows the filaments to function efficiently.

_____

(1 mark)

(iii) State ONE OTHER characteristic of the filaments that helps in their function.

_____

_____

(1 mark)

(b) (i) Identify TWO structures in the breathing apparatus of humans that have similar functions to the parts of the gills illustrated in Figure 5

Gill structure	Structure in the human breathing apparatus

(4 marks)

(c) State THREE ways in which the lungs of a smoker may be affected by the smoking habit.

_____

_____

_____

(3 marks)

(d) (i) Smoking is one human activity that pollutes the environment. Identify TWO OTHER human activities that result in pollution.

_____

_____

(2 marks)

(ii) Identify TWO effects of polluted waterways on aquatic life.

_____

_____

(2 marks)

Total 14 marks

END OF TEST